Aayush Shrivastava

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.92
Jeewan Public School	Class XII (CBSE)	2019	81.80%
Modern Public School	Class X (CBSE)	2017	9.80%

WORK EXPERIENCE

Research Assistant at Indian Institute of Technology Hyderabad

Guide: Dr Ramakrishna Upadrasta, Dept. of Computer Science and Engineering

- Specialization in the intersection of **Machine Learning** and **Systems**, actively addressing key challenges in this field as a part of my ongoing master's degree.
- Current projects focus on enhancing machine learning guided compiler optimizations by replacing heuristic-based optimization decisions with data-driven static cost models and using embeddings for automated program classification.

PROJECTS

ML-Guided Function Inlining Using Reinforcement Learning

- Implemented **Behavioural Cloning** into an existing RL-based inlining framework, logging per-callsite decisions to offline pretrain and warm-start the policy, reducing overexploration and improving overall performance by **3.56%**.
- Designed multiple cost model variants as an LLVM Module pass for reward computation, extracting basic-block sequences (peepholes) through DFS and random walks on the control-flow graph just before assembly generation.
- Incorporated BranchProbability weighting for each peephole using cumulative path probabilities derived from CFG edge weights, supporting both static (profile-free) and dynamic (profile-guided) cost estimation modes.
- Implemented call-graph frequency propagation via fixed-point iteration to estimate global function invocation counts from caller-callee block frequencies.

DocRAG: Intelligent PDF Chatbot Using Corrective RAG

- Developed an **AI-powered PDF chatbot** using **Corrective Retrieval-Augmented Generation (CRAG)** to deliver accurate, context-aware, and evidence-grounded responses to document queries.
- Engineered an **end-to-end CRAG pipeline** integrating document retrieval, relevance evaluation, conditional routing, query rewriting, web search, knowledge refinement and response generation using **LangGraph**.
- Implemented PDF ingestion, recursive text chunking, and **FAISS** vector indexing, leveraging **OpenAI's embedding model** for efficient semantic similarity search and retrieval across large documents.
- Improved **retrieval accuracy** and **response latency** through threshold-based relevance evaluation, persistent vector indexing and batched parallel LLM evaluations to minimize hallucinations and redundant computations.

Register Pressure Estimator at the LLVM IR level

- Designed and implemented **ComputeRegPressure**, an LLVM IR function pass that estimates register pressure and

spill count across a function using live-range tracking for each variable with loop-aware extension.

- Computed peak register pressure via a sweep-line algorithm over live intervals, with an optional BlockFrequency weighted mode for execution-aware estimates.
- Classified virtual registers into target register classes (GPR, FPR, VR) and simulated linear-scan register allocation to estimate spill count.
- Achieved **0.89 Spearman / 0.76 Kendall Tau** correlation for register pressure against MIR-level register pressure observed from LLVM's greedy register allocator.

Federated Uncertainty-Weighted Medical Image Segmentation

- Implemented a **CNN-based federated learning** framework for medical image segmentation with partially labeled and heterogeneous datasets, using a shared backbone and client-specific segmentation heads.
- Integrated **Monte Carlo Dropout**-based uncertainty estimation to reweight segmentation predictions and identify locally unknown anatomical structures across distributed clients.
- Achieved a mean Dice score of **84.77** on 2,413 3D medical images spanning 12 anatomical classes across 8 training and 2 test datasets, outperforming centralized averaging.

Program Classification using LLVM IR embeddings with Branch Probability Information

- Enhanced the IR2Vec embedding generation framework by integrating the **BranchProbabilityInformation (BPI)** pass from LLVM, enriching embeddings with control-flow likelihood information.
- Generated robust embeddings from source codes by converting to LLVM IR across all optimization levels (O0 to Oz), focusing on semantic similarity to classify and group programs.
- Implemented an **Multi Layer Perceptron (MLP)** model to classify programs using optimized LLVM IR embeddings and fine-tuned hyperparameters to maximize classification accuracy.
- Evaluated on POJ-104, CodeJam, and Codeforces datasets, achieving a **~6.5%** improvement in accuracy over baseline embeddings.

ForestBoost - Ensemble Learning Classifiers from Scratch

- Implemented a **Random Forest** Classifier from scratch, achieving **93.9%** test accuracy, and benchmarked performance against Scikit-learn's implementation (94.4%).
- Implemented a **Gradient Boosting** Classifier from scratch, tuned hyperparameters using GridSearchCV to achieve **99.8%** test accuracy, and benchmarked against a **Decision Tree** using Information Gain.
- Built an end-to-end **data preprocessing** pipeline incorporating missing-value imputation, feature selection, categorical encoding, PCA dimensionality reduction, and Min-Max scaling.
- Analysed the performance of ensemble models across varying hyperparameters, feature subsets and number of trees, evaluating accuracy, execution time, OOB error, and test error to study model sensitivity and generalization.

TECHNICAL SKILLS

- **Programming and Scripting Languages**: C/C++, Python, Java, SQL, Bash
- **Software Development Tools / Libraries**: LLVM, Git, MySQL, Docker, Vim, Visual Studio Code
- **Operating Systems**: Windows, Linux
- **Data Science & Machine Learning Tools / Libraries**: PyTorch, LangChain, OpenCV, Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, NLTK, Google Colab, Jupyter Notebook

- **Interests**: Software Development, Systems, Data Science, Generative AI, Retrieval-Augmented Generation (RAG), Deep Learning, NLP, Compiler Engineering

OPEN SOURCE CONTRIBUTIONS

- Authored and upstreamed 10+ PRs merged to the LLVM compiler infrastructure implementing compiler optimizations, backend codegen improvements, and toolchain bug fixes in C++.
- Implemented multiple IR-level simplification folds for comparison operations on extended operands (#218871), logical shift patterns (#200669), and floating-point sign/magnitude operations (#200836), reducing redundant instructions.
- Implemented an optimization folding signed max/min comparison patterns into equivalent bitwise shift forms, validated across AArch64, RISC-V, WebAssembly, and X86 backends. (#206242).

RELEVANT COURSES

- **Masters**: Foundations of Machine Learning, Deep Learning, Advanced DSA, Advanced Compiler Engineering, Program Analysis and Compiler Optimizations, Advanced Compiler Optimizations, Multimedia Content Analysis, Linear Optimization
- **Bachelors**: Data Structures and Algorithms, Operating Systems, Database Management Systems, Compiler Design, Computer Networks, Object Oriented Programming

ACHIEVEMENTS & CERTIFICATIONS

- Attained the **highest overall CGPA** in B.Tech (Computer Science and Engineering) for the session 2019-2023.
- Received a **Letter of Recommendation** for contributions on federated learning and uncertainty-aware medical image segmentation, recognising technical depth, originality, and research competence.
- Awarded the **Reliance Foundation Postgraduate Scholarship**, ranked among the top 100 PG students nationwide across 300+ institutions in India.
- Earned **Gold** Badge in Problem Solving, Java, and SQL on HackerRank for clearing the Intermediate level programming contests.
- Ranked **1307** out of 90,678 participants in the PrepSAT Hackathon, organized by PrepInsta.
- Represented as a **player** in the East Zone Inter-University Cricket Tournament organized by Ravenshaw University, Cuttack.
- Secured **Silver** Medal in the Inter-department Cricket Tournament (DIESTA) at IIT Hyderabad.
- Selected to represent IIT Hyderabad in the Group Dance at the **Inter-IIT Cultural Meet** held at IIT Patna.

POSITIONS OF RESPONSIBILITY

- **Lead Teaching Assistant** for CS3423 (Compilers-II) at IIT Hyderabad, responsible for setting exam questions, correcting assessments and designing the final project for the course.
- **Teaching Assistant** for CS6393 (Advanced Compiler Engineering) at IIT Hyderabad, designing assignments at regular intervals throughout the course and evaluating coding solutions.
- **Student Representative** for Master's students in the Senate Postgraduate Committee (SPGC) at IIT Hyderabad, acting as liaison between the AAC, DPGC, and CR Council to represent postgraduate students' academic concerns.
- **Placement Coordinator** in the Office of Career Services (OCS) at IIT Hyderabad, coordinating placement and internship processes between recruiters and UG/PG students.

- **PG Buddy** for Sunshine, the counselling cell at IIT Hyderabad, providing peer support and mental well-being outreach to students and connecting them with counselling resources and activities.

- **Hospitality Coordinator** for the 58th Inter-IIT Sports Meet 2025 at IIT Hyderabad, managing accommodation and hospitality logistics for 2,500+ students across 23 IITs.